Applying Maximum Entropy Models to Organism Sensors

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Motivation

How do biological organisms understand their enviornment?



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1. Encodes information about enviornment
2. Past, present, and future
3. Constraint: energetic cost of encoding



Entropy: a Fundamental Player

How well can we encode/compress information?

Shannon Entropy

For a random variable X and event space $\mathcal{X}$,

$$H[X] = -\sum_{x \in \mathcal{X}} p(x) \log_b(p(x)), \tag{1}$$

where $b = 2, e, 10$ most commonly.

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Key takeaways:

- $H[X, Y] \leq H[X] + H[Y]$, equality if X, Y ind.
- "Surprisal": $-\log p$
- Maximum entropy maximizes surprisal

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where S is the environment/stimulus and X is the biological sensor system.

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where S is the environment/stimulus and X is the biological sensor system.

- Typically used in the field
- Yields one loosely interpretable value given S, X
- Often difficult to calculate (many estimators are needed)
- Gives just one piece of information about sensor-environment system

Maximum Entropy: An Alternate Lens

The general pipeline

- Given constraints, get $p(\mathbf{x})$ that maximizes $H[X]$ (maximal 'surprisal').
  - $\rightarrow$ Least biased model
  - $\rightarrow$ Equivalent to minimizing KL divergence
  - $\rightarrow$ Stat mech models work too!

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$$p(\mathbf{x}) = \frac{1}{Z} \exp\left(\sum_{\mu} \lambda_\mu f_\mu(\mathbf{x}) \right) \tag{7}$$

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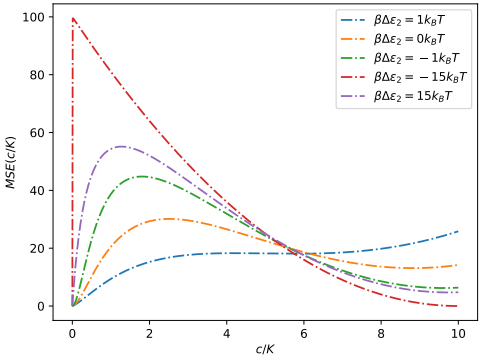
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- How good is the estimator?
  - → Ordered: bias, variance
  - → Categorical: confusion matrices

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MaxEnt: New Insights with Cooperative Sensors

- Two-sensor system: Cooperativity or not?
- Mutual information
  - → No cooperativity is best
- MaxEnt
  - → Depends on the ligand concentration
  - → More aligned with previous findings



MaxEnt: More Information with Cultured Cortical Neurons

- Model probability distribution ($\mathbf{x} = (x, \mathbf{s})$):

$$p(\mathbf{x}) = \frac{1}{Z} \exp\left(\mathbf{x}^\top J \mathbf{x}\right) \tag{8}$$

- Estimator: compute likelihood ratio via

$$L(\mathbf{s}) = \exp(-J_{xx} - 2J_{xs} \cdot \mathbf{s}) \tag{9}$$

	Actual 0	**Actual 1**
Predicted 0	True negative rate	False negative rate
Predicted 1	False positive rate	True positive rate

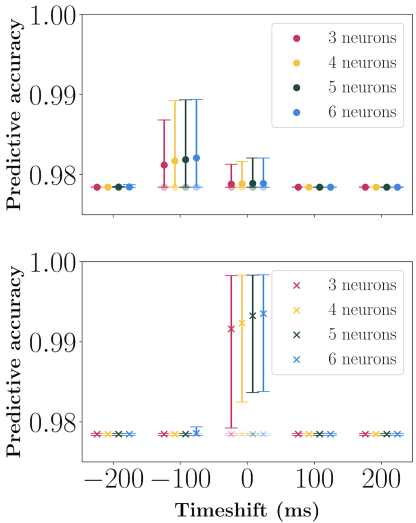
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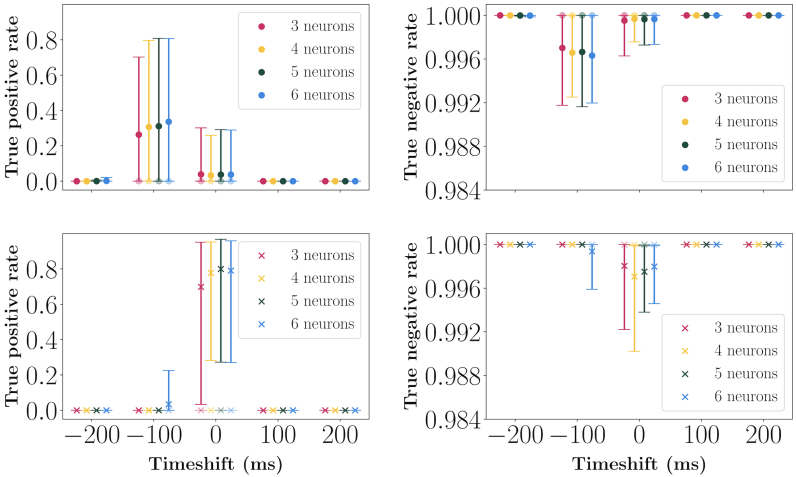
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Recap

	Mutual Information	Bias/Variance	Confusion Matrices
Variable Type	Any	Ordered / continuous	Categorical
Output	Single scalar	Bias, Variance as a function of input	Error rates per category
Interpretation	One loosely interpretable value	Accuracy (bias), precision (variance)	Per-class error breakdown
Computational Scaling	Exponential	Roughly power law	Power law or exponential, depending on situation

Acknowledgements

- PI: Sarah Marzen
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Thank you!

Here's the paper for more info: https://www.biorxiv.org/content/10.64898/2026.01.12.698979v1

